

EEU44C04 / CS4031 / CS7NS3 / EEP55C27

Next Generation Networks

Mobile architectures: LTE, LTE-A, 5G NR

三种移动通信技术:

LTE (Long Term Evolution): 通常说的 4G

LTE-A (LTE Advanced): LTE 的增强版

5G NR (New Radio): 5G 的无线接口

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Motivation 为什么要演进

- Evolution from 2G and 3G cellular systems
- Drivers
 - ✓ Exponential increase in demand
 - ✓ Support for higher data rates
 - ✓ Need for lower latency
 - ✓ Inter-networking with legacy network
 - ✓ Flexibility in spectrum use
 - ✓ Lower capital and operating expenses for network operator
- Smooth transition to 4G (IMT-Advanced)

2G、3G 已经不够用了，原因有：

用户需求指数级增长

需要更高的数据速率

需要更低的延迟

要能和旧网络互通（手机从 4G 掉到 3G 不能断）

频谱使用要灵活

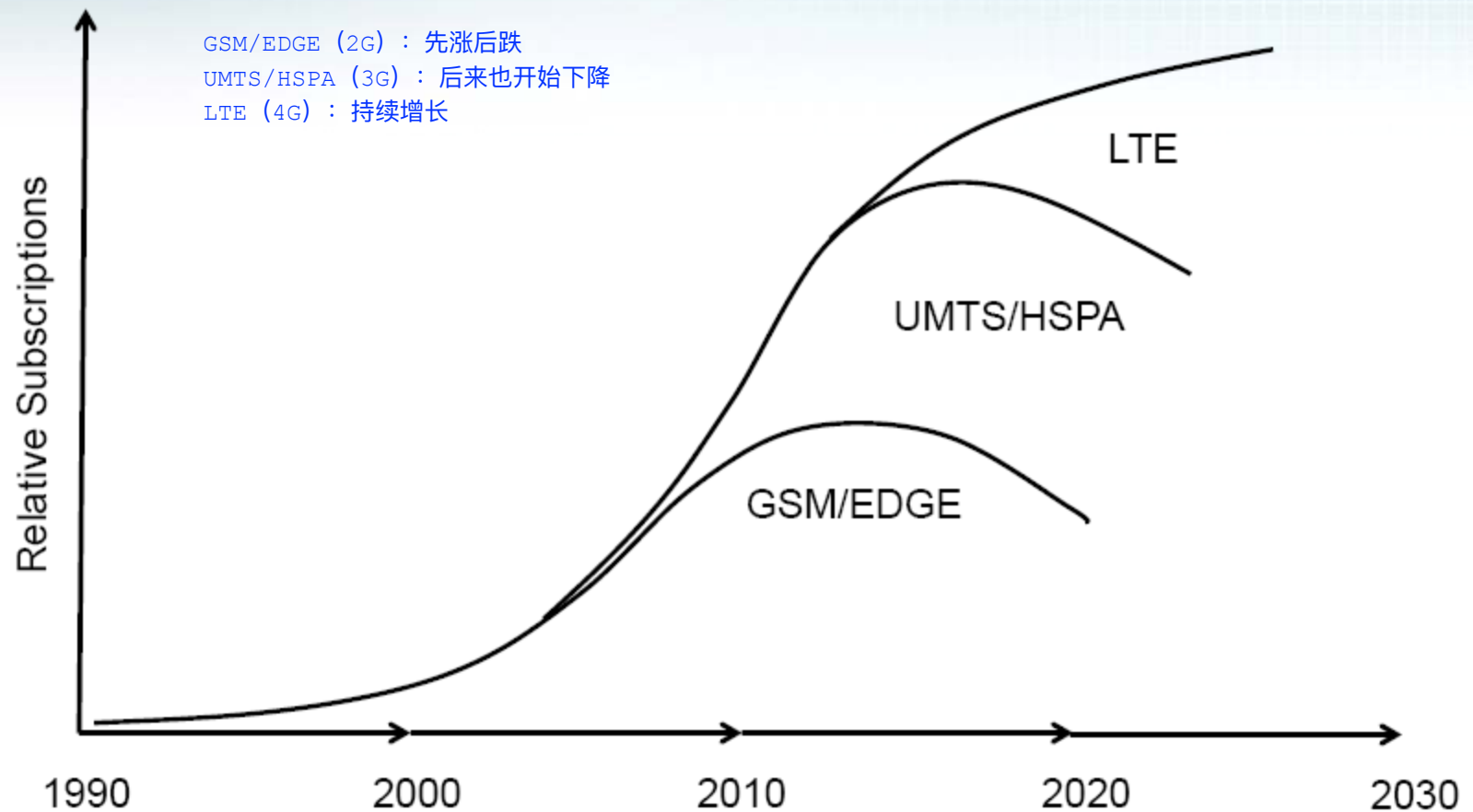
运营商要降低建设成本和运营成本

目标是平滑过渡到 4G (IMT-Advanced)，这是国际电信联盟 (ITU) 对"真正 4G"的正式定义。

Projected adoption

Projected adoption (预计的普及趋势)

这张图是 2010 年的预测，横轴是年份，纵轴是相对用户数：



Source: P. Rysavy, "Transition to 4G", 2010

LTE performance objectives

- 100 Mbps peak DL, 50 Mbps peak UL
 - ✓ An order of magnitude higher for LTE-A
 - ✓ Reduced latency (down to 10 ms)
 - ✓ Scalable bandwidth (up to 20 MHz)
- Backward compatibility with GSM/EDGE/UMTS
 - ✓ Handover and roaming to legacy networks
 - ✓ Use of existing spectrum
- Wide application
 - ✓ FDD (paired) and TDD (unpaired) operation
 - ✓ Mobility up to 350 km/h
 - ✓ Wide range of terminals: phones, smartphones, cameras, ...

DL = Down-Link

UL = Up-Link

TDD/FDD = Time/Frequency Division Duplexing

LTE performance objectives (LTE 性能目标)

峰值速率: 下行 100 Mbps, 上行 50 Mbps。LTE-A 要再高一个数量级

延迟低至 10 ms

带宽可伸缩, 最高 20 MHz

向后兼容 GSM/EDGE/UMTS: 能切换、漫游, 也能用现有频谱

应用广泛:

FDD (频分双工): 上下行用不同频率

TDD (时分双工): 上下行用同一频率、不同时间

支持移动速度最高 350 km/h (高铁)

终端类型多样 (手机、相机等)

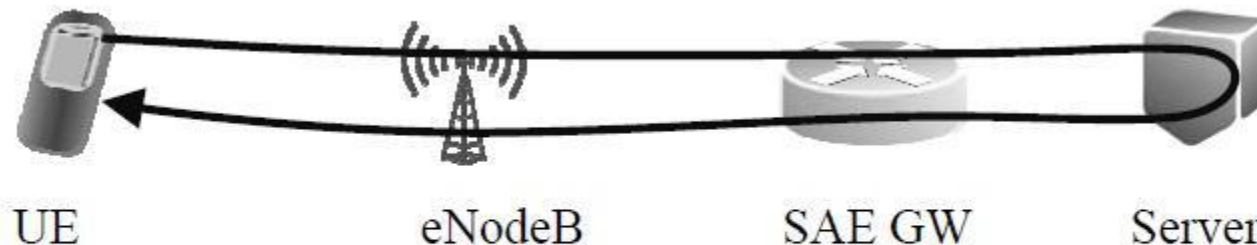
What is latency?

延迟是从你发出请求（下载文件、打开网页）到收到响应所经过的时间。

类比：带宽（bit rate）像水管有多粗，延迟像你拧开水龙头后第一滴水多久到。
有时延迟比速率更重要，比如打游戏、远程控制。

测量方法是测一个 IP 包往返的时间 (Round Trip Time): 手机 (UE) → 基站 (eNodeB) → 核心网网关 (SAE GW) → 服务器, 再原路返回。

- The time that passes from a user sending a piece of data request (e.g., file download or loading a webpage), to the time when the user gets a response
 - ✓ Sometimes more important than the bit rate offered
- The latency can be measured by the time it takes for an IP packet to travel from the terminal through the network to the Internet server, and back



Round trip time measurement

LTE Features

LTE Features (LTE 的主要特点)

OFDM/OFDMA: 把一个宽频带切成很多窄的子载波来传数据, 频谱效率更高, 所以速率更高

All-IP 架构: 全部基于 IP 传输, 网络管理更简单, CAPEX (建设成本) 和 OPEX (运营成本) 更低

频谱灵活: 能在不同频段、不同带宽工作, 还能把资源聚合起来用

- OFDM/OFDMA
 - ✓ Increased spectral efficiency → increased data rates
- All-IP architecture
 - ✓ Simplified network management
 - ✓ Reduced CAPEX and OPEX
- Spectrum flexibility
 - ✓ Operation in various bands, with variable bandwidth
 - ✓ Resource aggregation

OFDM = Orthogonal Frequency Division Multiplexing

正交频分复用

OFDMA = Orthogonal Frequency Division Multiple Access

正交频率划分多重访问

Spectrum flexibility

Spectrum flexibility (频谱灵活性)

可以在多种频谱上工作: 现有 3G 频谱、迁移过来的 2G 频谱 (如 900 MHz)、重新规划 (refarming) 的频谱 (如 UHF)

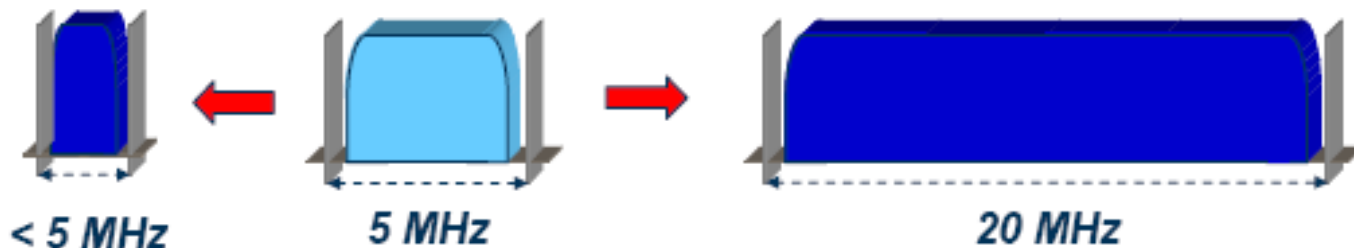
可以用不同大小的频带:

最高 20 MHz: 满足高速率

小于 5 MHz: 方便 2G/3G 老频谱逐步迁移过来 (旧频谱本来就窄)

图里是三个宽度不同的“块”, 代表 LTE 可以按需要选择带宽。

- Operation in a wide range of spectrum
 - ✓ Current 3G spectrum
 - ✓ Migration of 2G spectrum (e.g., 900 MHz)
 - ✓ Refarming of spectrum (e.g., UHF, 300 MHz–3 GHz)
- Operation in different-sized bands
 - ✓ Up to 20 MHz to support high data rates
 - ✓ Less than 5 MHz to enable smooth 2G & 3G spectrum migration



Architecture (LTE 网络架构)

eNB (Evolved Node B): LTE 基站

E-UTRAN: 所有 eNB 组成的无线接入网

MME (Mobility Management Entity): 管理用户的移动性、认证等控制信令

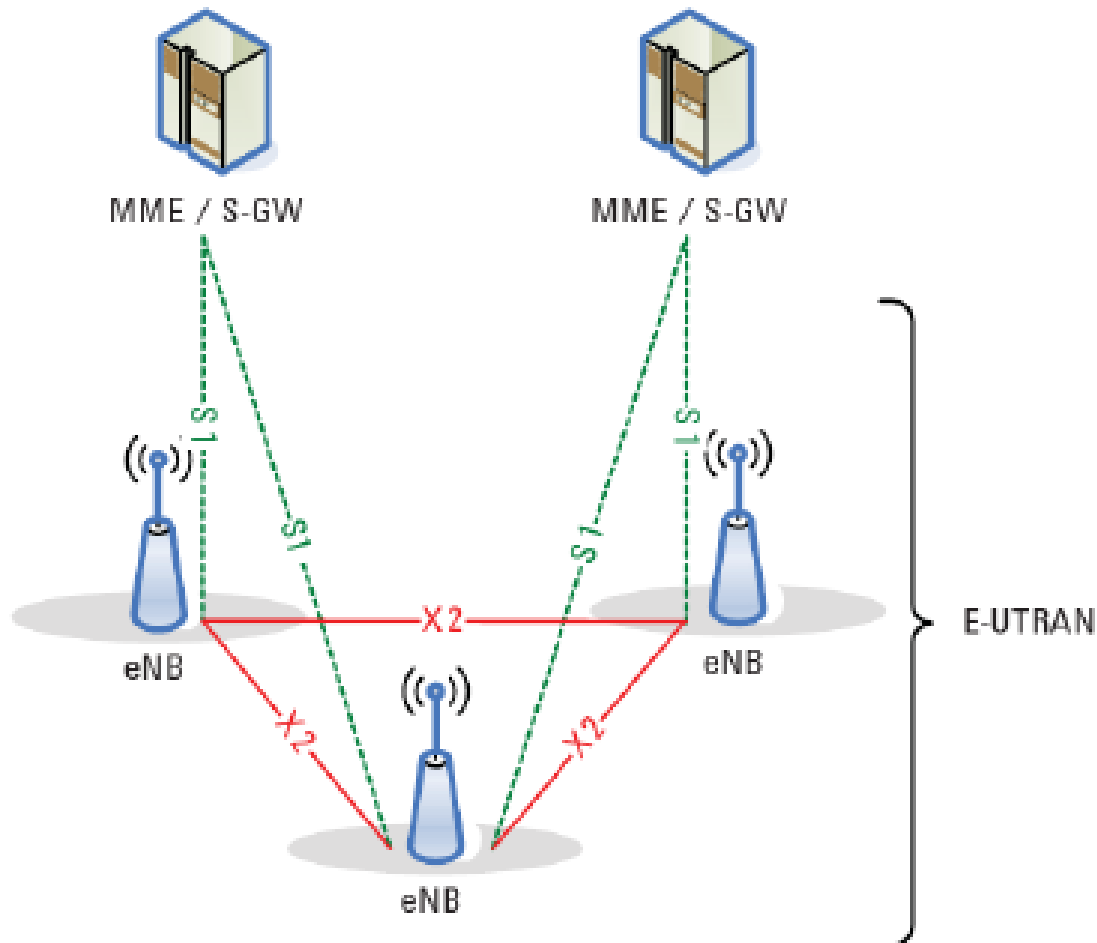
S-GW (Serving Gateway): 转发用户的数据包

S1 接口 (绿线): eNB 连接核心网 (MME/S-GW)

X2 接口 (红线): eNB 之间直接互联, 用于切换、干扰协调等

Architecture

(补充) 4G 把基站做得更“扁平”, 基站之间能直接通信, 减少延迟。



MME = Mobility
Management Entity

S-GW = Serving
Gateway

eNB = Evolved Node B

E-UTRAN = Evolved
Universal Terrestrial
Radio Access Network

进化的通用地面无线电接入网络

Protocol layers

- Packet data convergence protocol (PDCP) → Layer 2
 - ✓ IP header compression 数据包数据收敛协议
 - ✓ Cyphering and integrity protection
- Radio link control (RLC) → L2
 - ✓ Segmentation and concatenation
 - ✓ Retransmission handling
 - ✓ In-order delivery to higher layers
- Medium access control (MAC) → L2
 - ✓ Uplink and downlink scheduling at eNB
 - ✓ Hybrid Automatic Repeat ReQuest (HARQ)
- Physical layer (PHY) → L1
 - ✓ Coding/decoding
 - ✓ Modulation/demodulation (OFDM)
 - ✓ Multi-antenna use

Protocol layers (协议层)

层	属于	作用
PDCP	L2	IP 头压缩、加密 (ciphering)、完整性保护
RLC	L2	分段与拼接、重传处理、按顺序交给上层
MAC	L2	基站上下行调度、HARQ
PH	L1.	编码/解码、调制/解调 (OFDM)、多天线

PHY (物理层)

下行用 OFDMA, 上行用 SC-FDMA (都带循环前缀)。(补充) SC-FDMA 的信号峰均比更低, 更省手机电池

双工方式: FDD 和 TDD

无线帧 10 ms = 20 个时隙, 每个 0.5 ms

调制方式: QPSK、16-QAM、64-QAM

最大信息块 6144 bits, 用 CRC-24 检错

PHY

- OFDMA with cyclic prefix in DL, SC-FDMA with cyclic prefix in UL
- Duplexing: FDD and TDD
- 10 ms radio frames, containing 20 slots of 0.5 ms duration
- Supported modulation schemes: QPSK, 16-QAM, 64-QAM
 - ✓ Broadcast channel uses QPSK
- Maximum information block size: 6144 bits
- CRC-24 for error detection

为什么广播信道用 QPSK?

QPSK 每个符号只有 2 bit, 星座点之间距离大, 抗噪声能力最强。广播信息要让小区内所有人都收到, 包括信号很差的小区边缘用户, 而且广播是一对多, 没法按每个人的信道调整, 所以必须选最稳的调制。

Question: why does the broadcast channel use QPSK?

Scheduling

Scheduling (调度)

基站里的调度器把资源块 (resource block, 资源分配的最小单位) 在一定时间内分配给用户

一个时隙含 7 个 OFDM 符号 (短循环前缀) 或 6 个 (长循环前缀)

长循环前缀用于多径更严重的环境 (信号反射路径差异大)

子载波数量随带宽变化, 但子载波间隔固定

- Scheduler in eNB allocates resource blocks (which are the smallest elements of resource allocation) to users for predetermined amount of time
- Slots consist of either 6 (for long cyclic prefix) or 7 (for short cyclic prefix) OFDM symbols
- Longer cyclic prefixes are used to address longer multipath fading channel length
- Number of available subcarriers changes depending on transmission bandwidth
 - ✓ Subcarrier spacing is fixed

OFDMA (1)

OFDM 用大量窄子载波传输

下行物理资源可以看成一张时间-频率网格:

频率方向子载波间隔 $\Delta f = 15 \text{ kHz}$

符号时长 = $1/\Delta f + \text{循环前缀}$ (约 $66.7 \text{ }\mu\text{s} + \text{CP}$)

循环前缀 (CP) 的作用: 避免符号间干扰 (ISI), 还让接收机可以用最简单的单抽头均衡器

每个资源单元 (resource element) 承载一个 QPSK / 16QAM / 64QAM 符号, 例如 64QAM 每个资源单元 6 bit

- OFDM uses a large number of narrow sub-carriers for multi-carrier transmission
- The basic LTE downlink physical resource can be seen as a *time-frequency grid*
 - ✓ In the frequency domain, the spacing between the subcarriers is $\Delta f = 15\text{kHz}$
 - ✓ Symbol duration time: $1/\Delta f + \text{cyclic prefix}$
 - ✓ Cyclic prefix is used to avoid inter-symbol interference and allows the use of simpler receiver architectures (one-tap equalizer)
- One resource element carries QPSK, 16QAM or 64QAM
 - ✓ e.g., 64QAM: 6 bits per resource element

OFDMA (2)

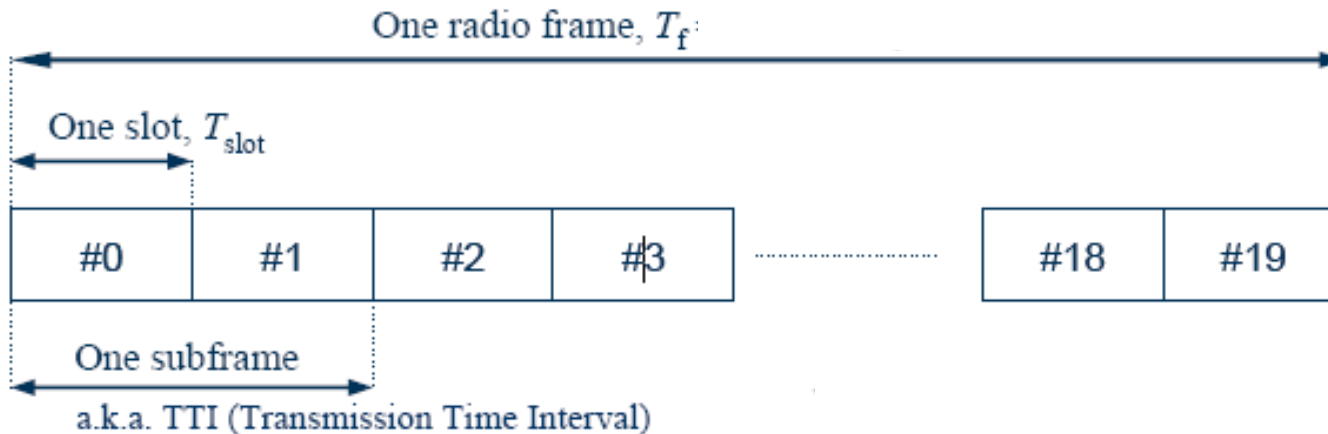
OFDM 符号组成资源块: 频域 180 kHz (12 个 $\times$ 15 kHz 子载波), 时域 0.5 ms (一个时隙)

TTI (传输时间间隔) = 1 ms = 2 个时隙 = 1 个子帧

一个无线帧 = 10 个子帧 = 20 个时隙 (#0 到 #19)

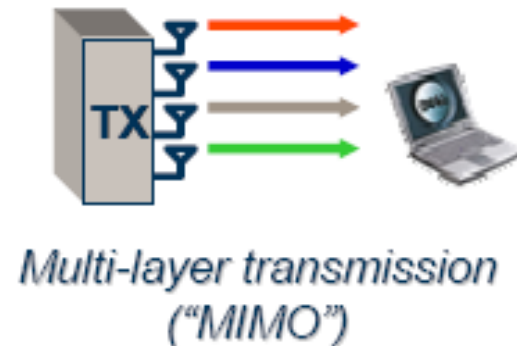
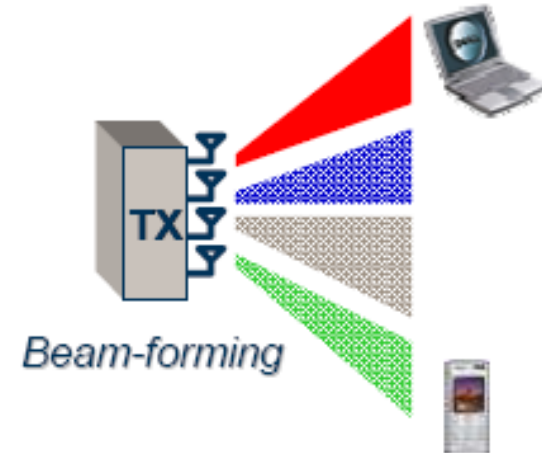
TTI 是调度的最小时间单位, 基站每 1 ms 就可以重新分配一次资源。

- OFDM symbols are grouped into resource blocks
 - ✓ Resource blocks have a total size of 180kHz in the frequency domain and 0.5ms in the time domain
 - ✓ Each 1ms Transmission Time Interval (TTI) consists of two slots (T_{slot})



Multi-Antenna techniques

- Transmit Diversity
- Beam-forming
- Spatial multiplexing (a.k.a. multi-layer transmission)
- Multi-user MIMO



Multi-Antenna techniques (多天线技术)

Transmit diversity (发射分集)：多根天线发同一信息，提高可靠性

Beam-forming (波束成形)：把信号能量集中指向某个用户，像手电筒

Spatial multiplexing (空间复用，也叫多层传输)：多根天线同时发不同数据流，提高速率

Multi-user MIMO：同一资源同时服务多个用户

MIMO = 多输入多输出 (多根发射天线、多根接收天线)。

MIMO = Multiple Input Multiple Output

LTE Advanced (LTE-A) : goals

- *Peak data rates*: DL - 1 Gbps; UL - 500 Mbps
- *Spectrum efficiency*: 3 times greater than LTE
- *Peak spectrum efficiency*: DL - 30 bps/Hz; UL - 15 bps/Hz
- *Spectrum use*: the ability to support scalable bandwidth use and spectrum aggregation (non-contiguous spectrum use)
- *Latency*: from Idle to Connected in < 50 ms; < 5 ms one way for individual packet transmission
- *More advanced MIMO*
- *Cell edge user throughput*: twice that of LTE
- *Average user throughput* : 3 times that of LTE
- *Mobility*: same as that in LTE
- *Compatibility*: LTE Advanced shall be capable of interworking with LTE and 3GPP legacy systems

LTE-A goals (LTE-A 目标)

峰值速率: 下行 1 Gbps, 上行 500 Mbps

频谱效率是 LTE 的 3 倍; 峰值频谱效率下行 30 bps/Hz, 上行 15 bps/Hz

支持可伸缩带宽和频谱聚合 (可以用不连续的频段)

延迟: 从空闲到连接 < 50 ms, 单包单向 < 5 ms

更先进的 MIMO

小区边缘用户吞吐量是 LTE 的 2 倍, 平均吞吐量是 3 倍

移动性同 LTE, 且兼容 LTE 和 3GPP 旧系统

Figure 1: Mobile subscriptions by technology

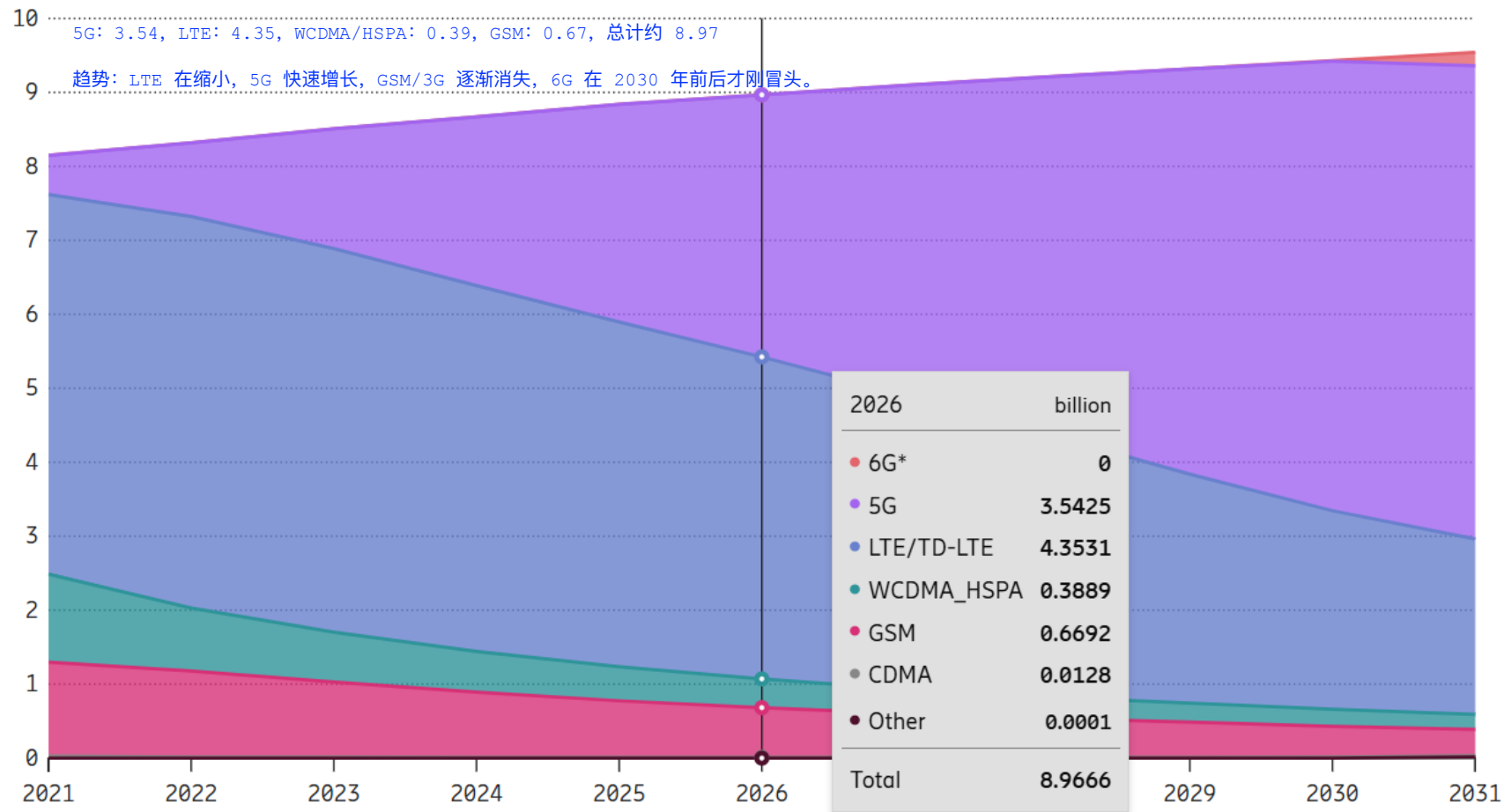


Mobile subscriptions by technology (爱立信订阅量图)

图展示 2021 到 2031 年各种技术的全球订阅数（十亿）。以 2026 年为例：

5G: 3.54, LTE: 4.35, WCDMA/HSPA: 0.39, GSM: 0.67, 总计约 8.97

趋势：LTE 在缩小，5G 快速增长，GSM/3G 逐渐消失，6G 在 2030 年前后才冒头。



*This does not include early uptake of subscriptions for AI-enabled devices such as autonomous vehicles, smartglasses and drones.

LTE-A Pro

LTE-A Pro 是 LTE-A 到 5G 之间的过渡, 目标是 Gbps 以上峰值速率、更均匀的体验、更好覆盖、更低延迟。四个关键点:

载波聚合演进: 聚合更多载波、不同类型频谱、不同小区

非授权频谱上的 LTE: 利用大量免费的非授权频谱

TDD/FDD 演进: 延迟更低, 上下行配置可自适应

更多天线: 走向 massive MIMO, FD-MIMO 用 3D 波束成形, 提高容量和覆盖

Propel mobile broadband even further

Enhance user experience and deliver efficient solutions to increase capacity



Gbps+ peak rates

More uniform experience

Better coverage

Significantly lower latencies

Carrier Aggregation evolution—wider bandwidths

Aggregating more carriers, diverse spectrum types and across different cells

LTE in unlicensed spectrum

Make the best use of the vast amounts of unlicensed spectrum available

TDD/FDD evolution—faster, more flexible

Enable significantly lower latency, adaptive UL/DL configuration, and more

Many more antennas—path to massive MIMO

Exploit 3D beamforming (FD-MIMO) to increase capacity and coverage

FD-MIMO = Full Dimension Multiple Input Multiple Output

LTE-A Pro

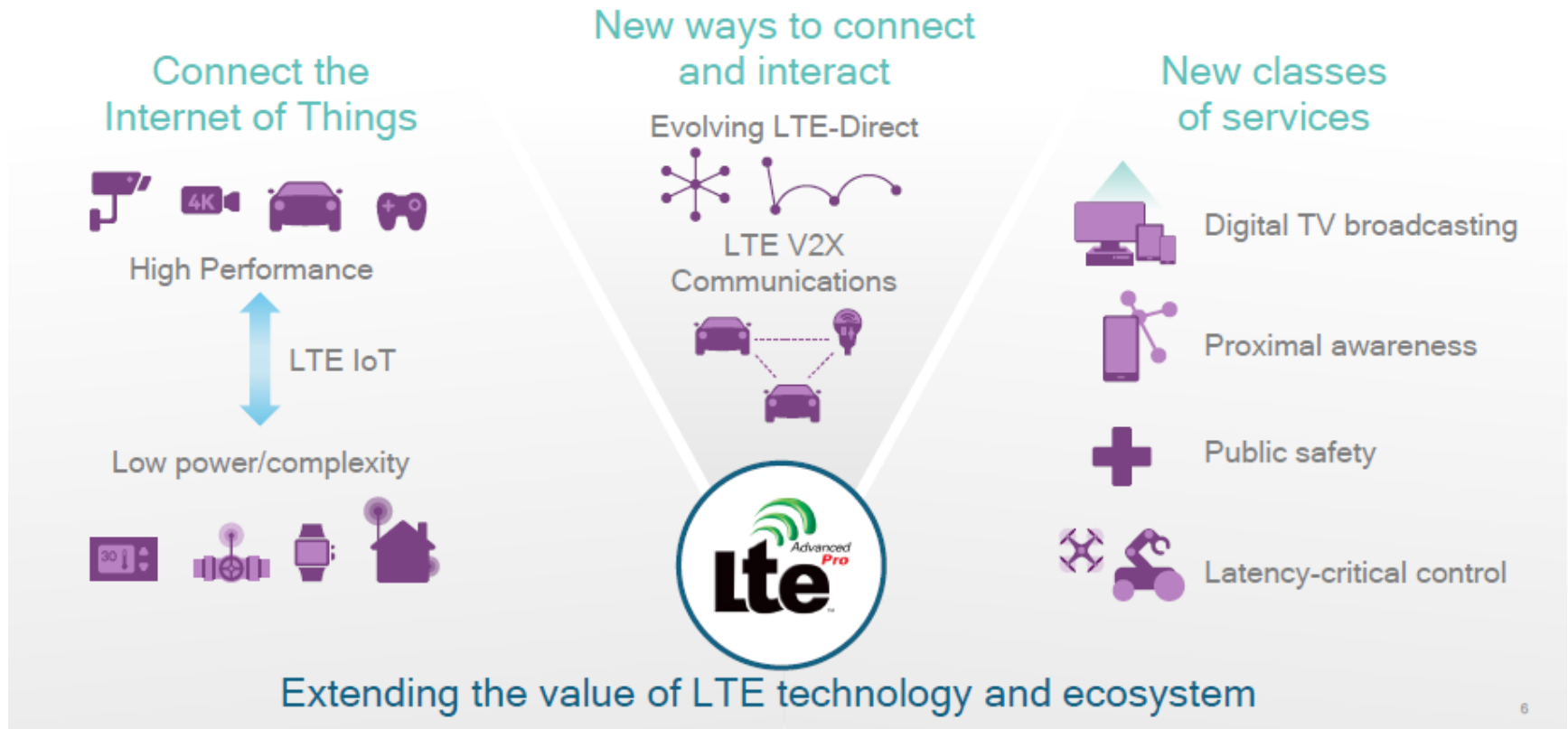
把 LTE 扩展到新用例：

物联网 (IoT)：从高性能设备（摄像头、汽车）到低功耗低复杂度设备（传感器、智能手表、智能家居）

新的连接方式：LTE-Direct（设备直连）、LTE V2X（车与一切通信）

新服务类型：数字电视广播、邻近感知、公共安全、低延迟关键控制（无人机、机器人）

Proliferate LTE to new use cases



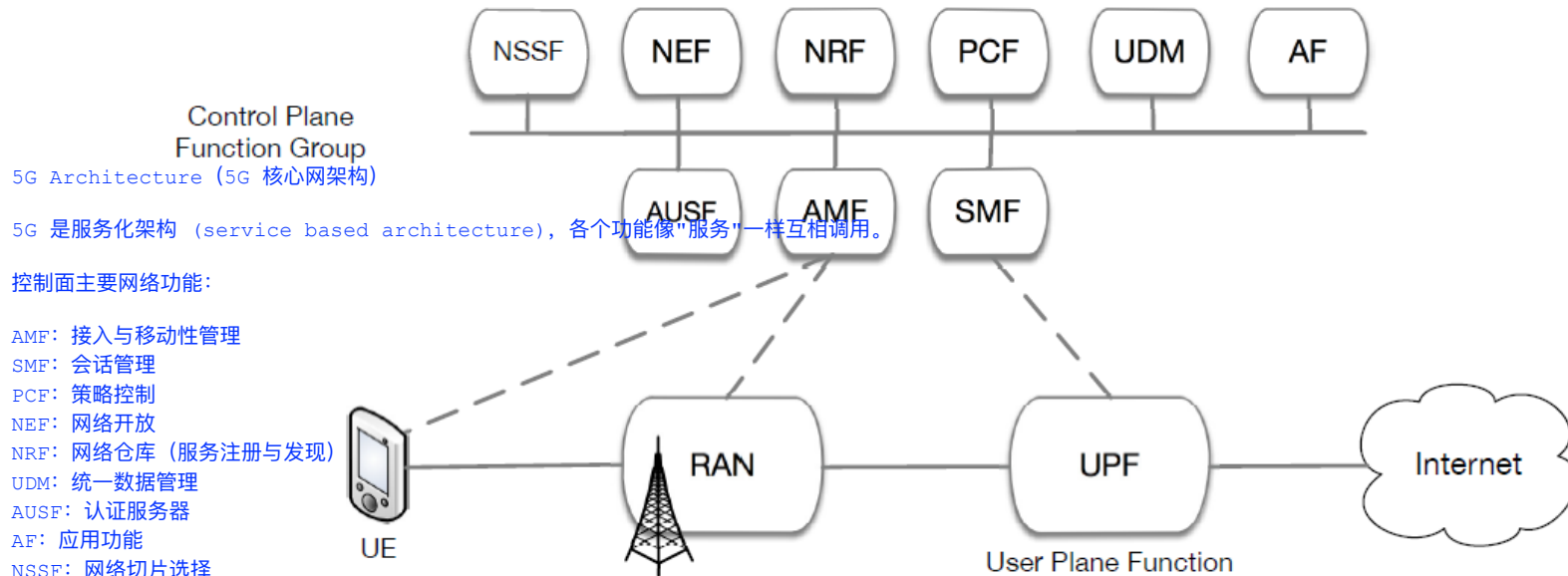
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V2X = Vehicle-to-everything

Qualcomm, "Leading the path towards 5G with LTE Advanced Pro", Jan 2016

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Architecture



用户面: 手机 (UE) → RAN (无线接入网) → UPF (用户面功能) → Internet。虚线是控制信令, 实线是用户数据。

(补充) 5G 把控制面和用户面分开, 还支持网络切片 (在同一张物理网络上划分出不同用途的虚拟网络)。

RAN = Radio Access Network

- A service based architecture.
- AMF : Access and Mobility, SMF : Session Mgt, PCF : Policy Control, NEF : Network Exposure, NRF : Network Repository, UDM : Unified Data Mgt, AUSF : Authentication Server, AF : Application Function, NSSF : Network Slice Selection
- For more information : [3GPP 23.501, 2018]

Frequency bands and channel bandwidths

Although the standard defined a very wide range of possible bands, some are of particular interest (european perspective) :

- 700 MHz with maximum bandwidth of 20 MHz for coverage and/or machine-type communications.
- 1400 MHz only for downlink as a supplementary carrier for enhancing capacity.
- 3.4-3.8 GHz with maximum bandwidth of 100 MHz for capacity. The "core" band for 5G in early phases.
- 26 GHz with maximum bandwidth of 400 MHz for hot spots and small cells.
- Later : 42 GHz and 66-71 GHz.

Below 6 GHz, around 1.2 GHz may be available for mobile communications ; above 6 GHz (26+41+66), around 11 GHz are available.

Frequency bands (5G 频段与带宽, 欧洲视角)

700 MHz, 最大 20 MHz: 覆盖广, 适合机器类通信

1400 MHz: 只用于下行, 作为补充载波提升容量

3.4-3.8 GHz, 最大 100 MHz: 早期 5G 的"核心"频段, 主打容量

26 GHz, 最大 400 MHz: 热点、小基站 (毫米波)

以后: 42 GHz、66-71 GHz

6 GHz 以下大约有 1.2 GHz 可用, 6 GHz 以上大约 11 GHz。

规律: 频率低, 覆盖远但带宽少; 频率高, 带宽大速率高但覆盖短。

Flexible numerology

New waveform = CP-OFDM with flexible subcarrier spacing (SCS) and symbol duration.

Flexible numerology (灵活参数集)

5G 的新波形是 CP-OFDM, 子载波间隔 (SCS) 和符号时长可变

4G 只有一种配置 (15 kHz), 5G 可以灵活配置

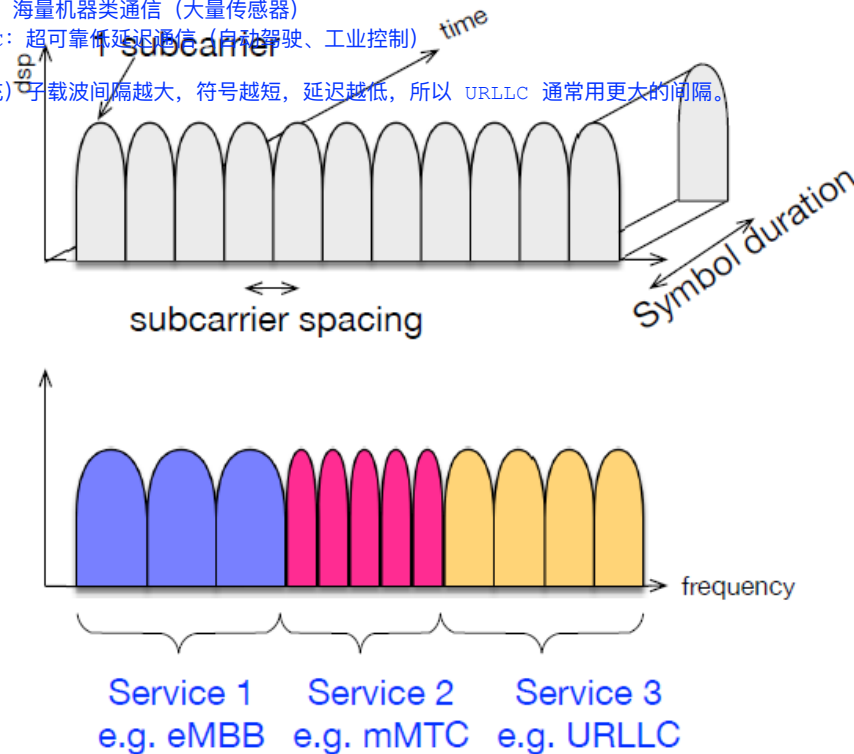
图中同一频带分给三种业务, 各用不同子载波间隔:

eMBB: 增强移动宽带 (高速上网、视频)

mMTC: 海量机器类通信 (大量传感器)

URLLC: 超可靠低延迟通信 (自动驾驶、工业控制)

(补充) 子载波间隔越大, 符号越短, 延迟越低, 所以 URLLC 通常用更大的间隔。



4G: A single configuration

5G: A flexible numerology

CP = Cyclic Prefix

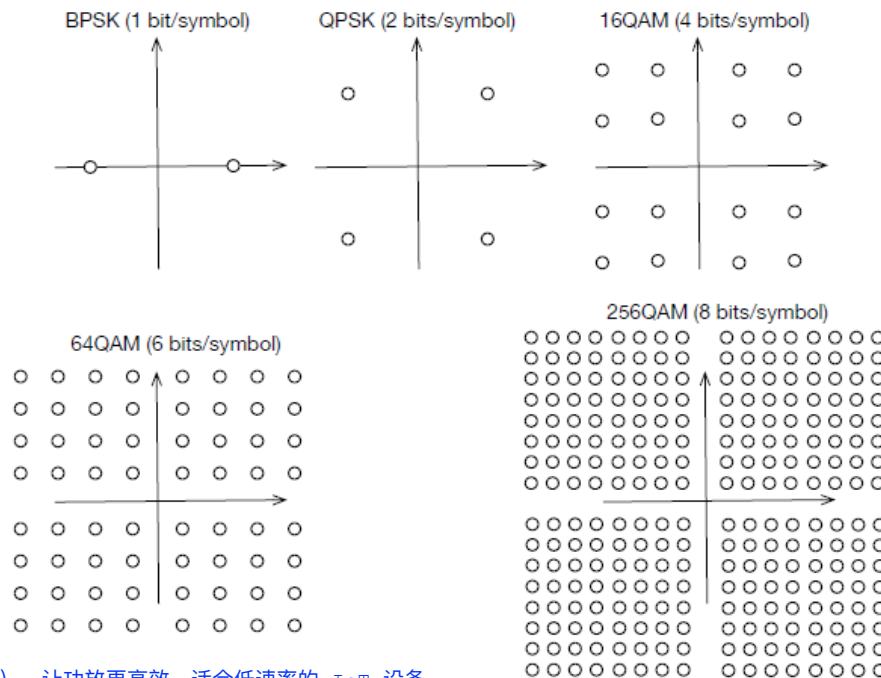
eMBB = enhanced Mobile BroadBand

mMTC = massive Machine Type Communication

URLLC = Ultra Reliable Low Latency Communication

Modulations

- **Modulations** : up to 256QAM (in both DL and UL); $\pi/2$ -BPSK in UL for reduced PAPR and better amplifier efficiency at low data rates for IoT.



Modulations (调制)

上下行最高支持 256QAM

上行的 $\pi/2$ -BPSK: 降低 PAPR (峰均功率比), 让功放更高效, 适合低速率的 IoT 设备

星座图每个点代表一个符号, 每符号比特数: BPSK 1、QPSK 2、16QAM 4、64QAM 6、256QAM 8。

规律: 点越多, 速率越高, 但点靠得越近, 越容易被噪声搞错, 所以只有信道好时才能用高阶调制。

For which of the following devices the battery lifetime is **not** a metric of major interest?

- ☐ Wireless sensors
- ☐ Mobile phones
- ☐ Cellular base stations

问：哪种设备不关心电池寿命？

答：蜂窝基站。基站有电缆供电，不像传感器和手机靠电池。



For which of the following devices the battery lifetime is NOT a metric of major interest?

- ☐ Wireless sensors
- ☐ Mobile phones
- ✓ Cellular base stations

Cellular base stations are generally cabled, so the power replenishment is not a major issue, as compared to wireless sensors and mobile phones

Which of these access techniques is the one adopted in 4th generation cellular systems (LTE/LTE-A) ?

- ☐ TDMA
- ☐ OFDMA
- ☐ CDMA



问：4G (LTE/LTE-A) 用哪种接入技术？

答：OFDMA。（补充记忆：2G 用 TDMA，3G 用 CDMA，4G 用 OFDMA。）

Which of these access techniques is the one adopted in 4th generation cellular systems (LTE/LTE-A) ?

- ☐ TDMA
- ☒ OFDMA
- ☐ CDMA

OFDMA is the correct answer. TDMA is used in 2G, and CDMA in 3G.

Given an OFDM transmission link in outer space, which of the following feedback schemes is the most efficient choice to provide channel state information to the transmitter?

- (i) The channel quality is reported for every subcarrier.
- (ii) The channel quality is reported for subchannels of N_s subcarriers, where N_s is greater than one and smaller than the maximum allowed number of subcarriers.
- (iii) The channel quality is reported only once, averaging it out over all the system bandwidth.



(iii) The channel quality is reported only once, averaging it out over all the system bandwidth.

→ in outer space there is no multipath (no scatterers) which means no frequency selectivity, therefore channel will have the same gain over all the bandwidth.

外太空 OFDM 链路

问：外太空的 OFDM 链路，向发射端反馈信道状态信息，哪种方式最高效？

- (i) 每个子载波都报告
- (ii) 每 N_s 个子载波一组报告
- (iii) 只报告一次，取整个带宽的平均

答：(iii)。外太空没有散射体，没有多径，信道没有频率选择性，所有频率的增益都一样，报告一次就够了，反馈开销最小。

(补充) 地球上有多径，不同频率的信道质量不同，所以实际用 (ii) 做折中。

A scheduler is a system component in charge of allocating over time the radio resources needed to transmit to the wireless users.

- A Round Robin (RR) schedules users in a round-robin fashion, i.e., cyclically. Another feature of RR, is that it also schedules users independently of their channel conditions.

Scheduler 与 Round Robin

调度器负责随时间把无线资源分给各个用户。

Round Robin (RR): 轮流调度, 每人一次, 循环进行, 完全不看信道好坏。



- A Maximum Carrier/Interference (Max C/I) schedules at time t the user $n_{\max}$ such that

$$n_{\max} = \arg \max_n \{d_n(t)\}$$

Where $d_n(t)$ is the expected throughput of user n at time t , if scheduled.

- A Proportional Fair (PF) schedules at time t the user $n_{\max}$ such that

$$n_{\max} = \arg \max_n \{d_n(t) / r_n(t)\} \quad (1)$$

Where $d_n(t)$ is the expected throughput of user n at time t , if scheduled, and $r_n(t)$ the average past throughput, i.e.,

$$r_n(t) = \overline{d_n(t')} \quad , \quad t' < t$$

Max C/I 与 Proportional Fair

Max C/I: 每次选当前预期吞吐量 $d_n(t)$ 最大的用户

$n_{\max} = \arg \max_n d_n(t)$

Proportional Fair (PF): 每次选 $d_n(t) / r_n(t)$ 最大的用户, 其中 $r_n(t)$ 是该用户过去的平均吞吐量

PF 的逻辑: 过去拿得多, r 大, 分数变低; 过去拿得少, r 小, 分数变高。这样既照顾信道好的用户, 也不会让差用户一直饿着。



a) Which of the three above schedulers maximizes fairness in resources allocation? Which maximizes throughput? Which achieves a trade-off between fairness and throughput? 哪个最公平? 哪个吞吐量最大? 哪个是折中?

b) Suppose we are given the following throughput traces for three users U_1 , U_2 , U_3 at times $t-2$, $t-1$, and the expected throughput at time t .

	$d_n(t-2)$ [Mbps]	$d_n(t-1)$ [Mbps]	$d_n(t)$ [Mbps]
U_1	0.5	1	0.8
U_2	2	1.7	1.5
U_3	0.6	1.1	2.1



Which users are going to be scheduled by RR, PF and Max C/I at time t ? 问 RR、PF、Max C/I 在时刻 t 各调度谁?

- a) - RR maximizes fairness.
- Max C/I maximizes throughput.
- PF achieves a trade-off, as its scheduling metric (see eq. (1)) takes into account both the throughput, and what happened in the past, equalizing the allocation of resources
- ✓ if high allocation (and therefore high throughput) in the past, scheduling metric decreases
 - ✓ if low allocation in the past, scheduling metric increases

(a) 的答案

RR 最公平 (人人轮流)

Max C/I 吞吐量最大 (总选信道最好的)

PF 是折中, 因为它同时考虑当前吞吐量和历史: 过去分配多, 指标下降; 过去分配少, 指标上升

- b) - RR allocates blindly and cyclically.
 Therefore if at $(t-2, t-1)$ allocation was,
 e.g., (U_1, U_2) at t RR will schedule U_3 . Any
 other scheduled users' triple is equally
 valid.
- Max C/I will allocate U_3 at t , since $d_3(t)=2.1$
 $> d_2(t)=1.5 > d_1(t)=0.8$
 - PF will also allocate U_3 , since

$$\begin{aligned}
 d_3(t)/r_3(t) &= 2.1/[(0.6+1.1)/2] \approx 2.47 \\
 &> d_1(t)/r_1(t) &= 0.8/[(0.5+1)/2] \approx 1.07 \\
 &> d_2(t)/r_2(t) &= 1.5/[(2+1.7)/2] \approx 0.81
 \end{aligned}$$

(b) 的答案

RR: 盲目循环。比如之前是 (U_1, U_2) , 这次就轮到 U_3 (其他组合同样合理, 因为题目没给之前的调度顺序)

Max C/I: 比 $d(t)$, $2.1 > 1.5 > 0.8$, 选 U_3

PF: 算 $d(t) / r(t)$, 其中 r 是前两次的平均:

$U_3: 2.1 / [(0.6+1.1)/2] = 2.1/0.85 \approx 2.47$

$U_1: 0.8 / [(0.5+1)/2] = 0.8/0.75 \approx 1.07$

$U_2: 1.5 / [(2+1.7)/2] = 1.5/1.85 \approx 0.81$

所以 PF 也选 U_3 。

这道题三种方法结果都是 U_3 , 只是巧合。实际中它们经常选不同的人。